ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 5

NATURAL SCIENCES AND TECHNOLOGY

PAPER 1 · PRACTICE B

MARKS: 60 **TIME: 1¼ hours**

An end-of-year test on Grade 5 Terms 3 and 4: Energy and Change (fuels and burning, renewable energy, electricity and saving it, energy and movement, wheels and axles) and Planet Earth and Beyond (the Earth, rocks, soil, and fossils).

Written by Academic Central for practice, following the CAPS curriculum for Grade 5. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Number your answers exactly as the questions are numbered.
3. Read each question carefully before you answer.
4. Write neatly and in full sentences where you are asked to explain.

QUESTION 1: MULTIPLE CHOICE

Choose the correct answer. Write only the letter (A–D) next to the question number, e.g. 1.11 B.

1.1 Which of these is a renewable source of energy?

A Coal B Petrol C Wind D Paraffin

1.2 Which of these is NOT a fuel?

A Wood B Coal C Water D Gas

1.3 Coal is found …

A deep under the ground. B inside trees.
C in the sea. D in the air.

1.4 When a match burns, its stored energy changes into …

A heat and light. B sound only.
C electricity. D food energy.

1.5 Which appliance changes electricity into movement?

A A kettle B A fan C A light bulb D A heater

1.6 Electricity reaches our homes through …

A water pipes. B power lines (cables).
C rivers. D roads.

1.7 Rock that has been changed by great heat and pressure deep under the ground is called …

A sedimentary rock. B igneous rock.
C metamorphic rock. D a fossil.

1.8 Rocks are slowly broken down into small pieces by wind, water, plants and changes in temperature. This is called …

A weathering. B photosynthesis.
C evaporation. D recycling.

1.9 Which type of soil holds the most water?

A Sand B Clay C Loam D Gravel

1.10 Fossils are most often found in …

A igneous rock.
B sedimentary rock.
C fresh lava.
D sea water.

[10]

QUESTION 2: TRUE OR FALSE

Write TRUE or FALSE next to the question number.

2.1 A candle needs air (oxygen) to keep burning.

2.2 Solar energy is energy from the Sun.

2.3 Coal will never run out.

2.4 It is safe to touch a plug with wet hands.

2.5 Clay soil is made of very small particles.

2.6 Humus is made from dead plants and animals that have rotted.

[6]

QUESTION 3: FUELS AND BURNING

Learners lit three candles of the same size. They left candle A open and covered candles B and C with glass jars, as shown in FIGURE 1.

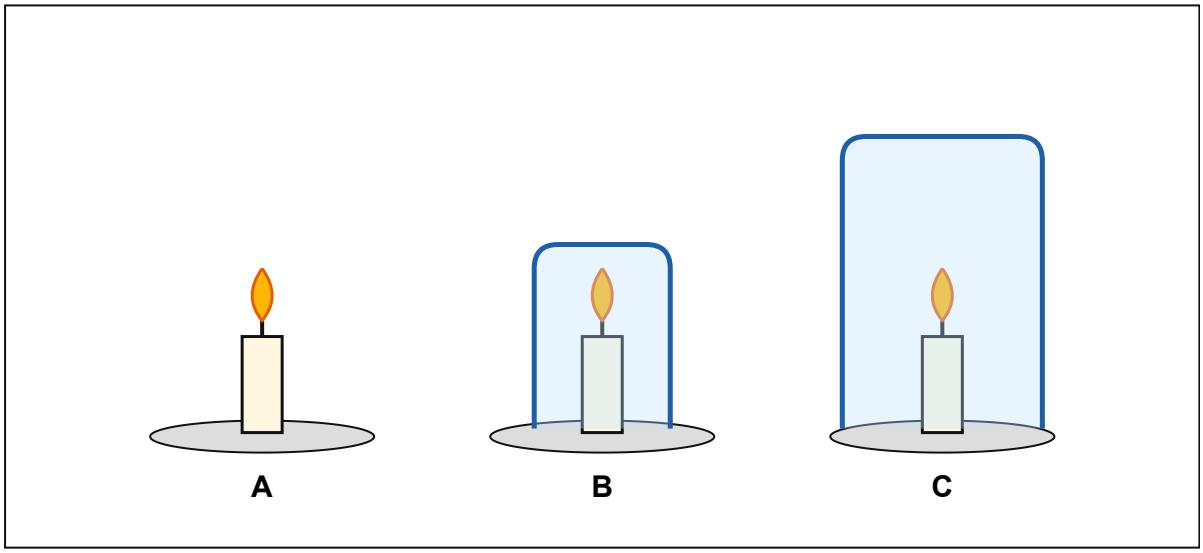


FIGURE 1: Three burning candles

3.1 Which candle will burn the longest? (1)

3.2 Which candle under a jar will go out first? Explain why. (2)

3.3 What does this investigation show that a fire needs? (1)

3.4 Oil in a pot on the stove catches fire. How can you safely put out the fire? (1)

3.5 Name TWO renewable sources of energy. (2)

3.6 Most of South Africa's electricity is made by burning coal. Give TWO reasons why we should save electricity. (2)

3.7 Name ONE fuel that a car uses. (1)

[10]

QUESTION 4: ELECTRICITY AND MOVEMENT

4.1 Complete the energy chain: Sun → plants that grew long ago → coal → power station → _____ → light bulb in your home (1)

4.2 Name TWO appliances that change electricity into heat. (2)

4.3 Give ONE advantage and ONE disadvantage of using cells (batteries) instead of mains electricity. (2)

4.4 Why should old cells not be thrown into the dustbin? (1)

4.5 Some people connect wires to the power lines without permission (illegal connections). Give TWO reasons why this is dangerous. (2)

4.6 Answer these questions about wheels and axles.

4.6.1 What is an axle? (1)

4.6.2 Why must wheels be round? (1)

4.6.3 Name TWO ways to give a toy cart energy so that it moves. (2)

4.6.4 Why do people put oil or grease on axles? (1)

4.7 What kind of energy does a stretched elastic band have? (1)

4.8 You wind up a toy car and let it go. Name the input energy and the output energy of this system. (2)

[16]

QUESTION 5: PLANET EARTH: ROCKS, SOIL AND FOSSILS

Learners put the same amount of three different soils, A, B and C, into funnels. They poured 100 ml of water onto each soil. FIGURE 2 shows how much water ran through each soil after 10 minutes.

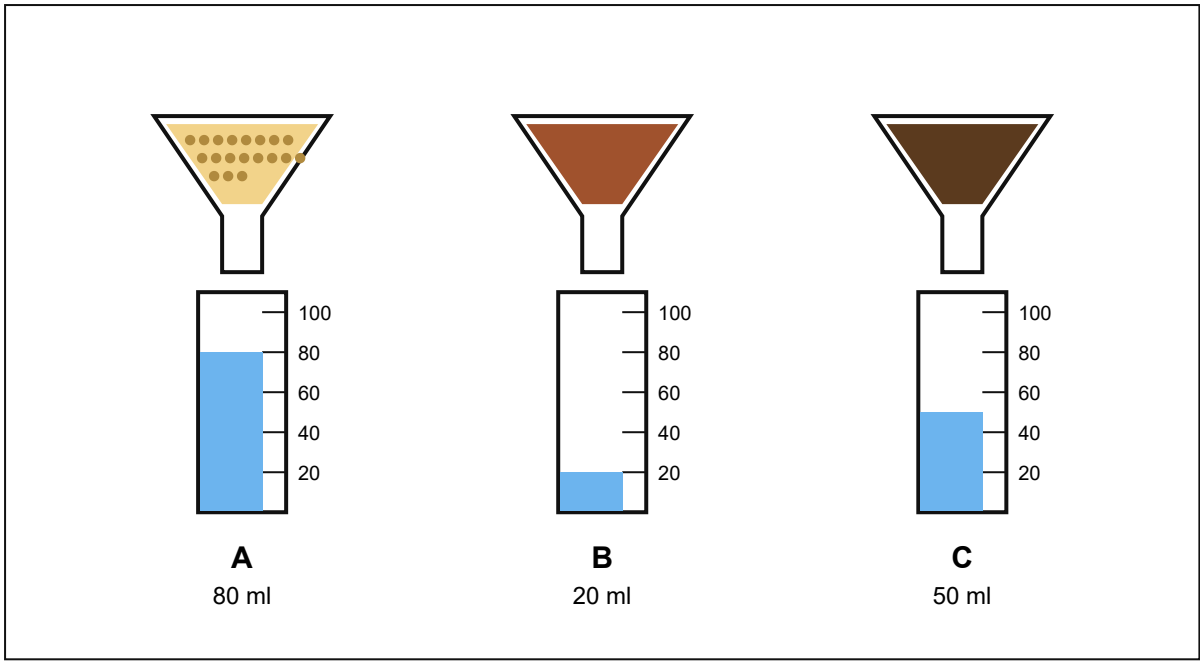


FIGURE 2: How much water ran through each soil

5.1	Which soil let the most water run through?	(1)
5.2	Soil A is sandy soil. Which soil, A, B or C, is most likely clay soil? Give a reason.	(2)
5.3	How much water did soil B hold back?	(1)
5.4	Name TWO things the learners had to keep the same to make this a fair test.	(2)
5.5	Why do farmers add compost to sandy soil?	(1)
5.6	Give TWO ways to stop soil from being washed away (soil erosion).	(2)
5.7	Give ONE example of an igneous rock and ONE example of a sedimentary rock.	(2)
5.8	Name TWO ways in which people use rocks.	(2)
5.9	Why are fossils important to scientists? Give TWO reasons.	(2)
5.10	A footprint of a dinosaur that has turned into rock is called a trace fossil. Name ONE other trace fossil.	(1)
5.11	Life is found only on Earth in our solar system. Give TWO reasons why plants and animals can live on Earth.	(2)

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TOTAL: 60

END OF QUESTION PAPER